# FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Chemistry — Class IX (SSC-I)

**Syllabus: Ch 1 (Nature of Chemistry) | Ch 2 (Matter) | Ch 3 (Atomic Structure) | Ch 4 (Periodic Table) | Ch 5 (Chemical Bonding) | Ch 6 (Stoichiometry) | Ch 7 (Electrochemistry) | Ch 8 (Energetics) | Ch 9 (Chemical Equilibrium)**

**ANSWER KEY (Version 3)**

| **Total Marks: 65** | **Time Allowed: 2h 30m** | **Version: 3** |
|---|---|---|

## SECTION A — MCQ Answer Key (12 × 1 = 12 Marks)

**Q1.** Branch studying heat changes and rates of reactions: **√ A — Physical chemistry** **1 mark**

**Physical chemistry** deals with the relationship between the physical properties of matter and chemical changes, including energy/heat changes and reaction rates (kinetics). Organic (B) studies carbon compounds; analytical (C) determines composition; nuclear (D) studies changes in the nucleus.

**Q2.** Direct change of gas to solid: **√ B — Deposition** **1 mark**

**Deposition** is the direct gas → solid change (e.g. water vapour forming frost). Sublimation (A) is the reverse (solid → gas). Condensation (C) is gas → liquid; freezing (D) is liquid → solid.

**Q3.** Maximum electrons in a *d* sub-shell: **√ C — 10** **1 mark**

Sub-shell capacities are: **s = 2, p = 6, d = 10, f = 14**. A *d* sub-shell has 5 orbitals × 2 electrons = **10 electrons**.

**Q4.** Electronic configuration of $Cl^-$ (Z = 17): **√ D — 2, 8, 8** **1 mark**

A neutral Cl atom (Z = 17) is 2, 8, 7. The chloride ion $Cl^-$ has **gained 1 electron** (18 electrons total), giving **2, 8, 8** — the stable argon configuration. Option A (2, 8, 7) is the neutral atom.

**Q5.** Largest atomic size: **√ C — K** **1 mark**

H, Li, Na and K are all in Group 1. Atomic size **increases down a group** as new shells are added. K (Period 4) has the most shells, so it is the largest: K > Na > Li > H.

**Q6.** Inert element with a completely filled outer shell: **√ D — Argon** **1 mark**

**Argon** (2, 8, 8) is a noble gas with a complete outermost shell, so it has no tendency to gain, lose or share electrons and is chemically inert. Sodium, chlorine and oxygen all have incomplete outer shells and are reactive.

**Q7.** Substance with a metallic bond: **√ A — Copper** **1 mark**

**Copper** is a metal; its atoms are held by a **metallic bond** (a lattice of positive ions in a "sea" of delocalised electrons). NaCl has ionic bonds, diamond has covalent bonds, and HCl has a (polar) covalent bond.

**Q8.** Mass of 0.25 mol $CaCO_3$: **√ B — 25 g** **1 mark**

Molar mass of $CaCO_3$ = 40 + 12 + 3(16) = **100 g/mol**. Mass = moles × molar mass = 0.25 × 100 = **25 g**.

**Q9.** Oxygen atoms in one mole of $CO_2$: **√ D — $1.204\times10^{24}$** **1 mark**

1 mole of $CO_2$ = $6.022\times10^{23}$ molecules. Each molecule has **2 O atoms**, so O atoms = 2 × $6.022\times10^{23}$ = $\mathbf{1.204\times10^{24}}$. Option A counts molecules, not O atoms.

**Q10.** Oxidation state of Cl in $ClO_3^-$: **√ A — +5** **1 mark**

In $ClO_3^-$, sum of oxidation states = −1 (the ion charge). O = −2, so 3 O = −6. Let Cl = x: x + (−6) = −1 → x = **+5**.

**Q11.** Minimum energy to start a reaction: √ **B — Activation energy** **1 mark**

**Activation energy ($E_a$)** is the minimum energy colliding molecules must have for an effective collision that starts the reaction. Bond energy is the energy to break one mole of a particular bond; enthalpy change is the overall heat change of the reaction.

**Q12.** A reversible reaction reaches equilibrium only in: √ **C — a closed system** **1 mark**

Equilibrium requires a **closed system** so that no reactant or product can escape; both forward and reverse reactions can then continue until their rates become equal. In an open container, products escape and equilibrium is never established.

## SECTION B — Short Questions Answer Key (11 × 3 = 33 Marks)

**Q2(i). Define analytical chemistry and physical chemistry; state what each studies.** **3 marks**

**Analytical chemistry:** the branch of chemistry that deals with the **qualitative and quantitative analysis** of substances — finding out *what* a sample is made of and *how much* of each component is present. **1½**
**Physical chemistry:** the branch that deals with the relationship between the **physical properties of matter and chemical changes**, including the energy (heat) changes and rates of chemical reactions. **1½**

### OR — Nuclear chemistry; one beneficial + one harmful use.

**Nuclear chemistry** is the branch that deals with changes taking place in the **nucleus of an atom** (radioactivity and nuclear reactions). **1**
**Beneficial use:** generation of electricity in nuclear power plants; use of radioisotopes in medicine (radiotherapy for cancer, diagnostic tracers). **1**
**Harmful effect:** harmful radiation that damages living cells / nuclear weapons cause mass destruction / disposal of radioactive waste pollutes the environment. **1**

**Q2(ii). Differentiate homogeneous and heterogeneous mixtures, with examples.** **3 marks**

| Homogeneous mixture | Heterogeneous mixture |
|---|---|
| Uniform composition throughout; single phase. | Non-uniform composition; two or more phases. |
| Components cannot be distinguished by the eye. | Components can be seen/distinguished separately. |
| **Example:** salt dissolved in water; air. | **Example:** sand in water; a mixture of iron filings and sulphur. |

**Any 2 differences = 2** **1 example each = 1**

### OR — What is an alloy? Example + constituent metals; why a solid solution.

An **alloy** is a homogeneous mixture (solid solution) of two or more metals, or of a metal with a small amount of a non-metal. **1**
**Example:** brass = copper + zinc (or bronze = copper + tin; steel = iron + carbon). **1**
It is a **solid solution** because one solid metal is dissolved uniformly (evenly distributed) within another solid metal, giving a single homogeneous solid. **1**

**Q2(iii). Maximum electrons in K, L, M shells and the rule.** 3 marks

The maximum number of electrons a shell can hold is given by the rule **$2n^2$**, where n = shell number. 1

```
K shell (n = 1): 2 × 1² = 2 electrons
L shell (n = 2): 2 × 2² = 8 electrons
M shell (n = 3): 2 × 3² = 18 electrons
```

**K = 2, L = 8, M = 18 → 2**

### OR — Define valence shell & valence electrons; valence electrons of S (Z = 16).

**Valence shell:** the outermost occupied electron shell of an atom. 1
**Valence electrons:** the electrons present in the valence (outermost) shell; they take part in chemical bonding. 1
Sulphur (Z = 16) has configuration **2, 8, 6**, so it has **6 valence electrons**. 1

**Q2(iv). Define electron affinity; variation across a period and down a group.** 3 marks

**Electron affinity:** the amount of energy released when an electron is added to a neutral, gaseous atom to form a negative ion. 1
(a) **Across a period (left → right):** electron affinity generally **increases** — atomic size decreases and nuclear charge increases, so the added electron is attracted more strongly and more energy is released. 1
(b) **Down a group:** electron affinity generally **decreases** — atomic size increases, the added electron is farther from the nucleus, so less energy is released. 1

### OR — Periodicity of properties; two properties that vary periodically.

**Periodicity of properties:** the regular, repeating pattern (recurrence) in the physical and chemical properties of elements when they are arranged in order of **increasing atomic number**. 1½
**Any two:** atomic size (radius), ionization energy, electron affinity, electronegativity, metallic/non-metallic character. 1½

**Q2(v). Coordinate covalent bond; formation in $H_3O^+$.** 3 marks

A **coordinate covalent (dative) bond** is a covalent bond in which the shared pair of electrons is **donated by only one** of the two bonded atoms (the donor); the other atom (acceptor) contributes no electron. 1
**Formation of $H_3O^+$:** an oxygen atom in a water molecule has two lone pairs of electrons. It **donates one lone pair** to a bare hydrogen ion $H^+$ (which has an empty orbital and no electrons). 1
The O atom is the **donor** and $H^+$ is the **acceptor**; this forms the hydronium ion $H_3O^+$, which carries one coordinate covalent bond. 1

### OR — Formation of MgO by electron transfer.

Mg (2, 8, 2) has 2 valence electrons; O (2, 6) needs 2 electrons to complete its octet. 1
Mg **loses 2 electrons** → $Mg^{2+}$ (2, 8, neon config); O **gains those 2 electrons** → $O^{2-}$ (2, 8, neon config). 1
The oppositely charged $Mg^{2+}$ and $O^{2-}$ ions attract by a strong electrostatic force = an **ionic bond**, forming MgO. 1

**Q2(vi). Define molar mass; molar mass of $Na_2CO_3$.** **3 marks**

**Molar mass:** the mass in grams of one mole of a substance; it is numerically equal to the formula/molecular mass (in amu) expressed in g/mol. **1**

```
Na2CO3: 2(Na) = 2 × 23 = 46
           C  = 12
        3(O) = 3 × 16 = 48
Molar mass = 46 + 12 + 48 = 106 g/mol
```

**Correct working = 1** **106 g/mol = 1**

### OR — Avogadro's number; atoms in 0.5 mol $N_2$.

**Avogadro's number = $6.022\times10^{23}$** = the number of representative particles in one mole of a substance. **1**

```
Molecules in 0.5 mol N2 = 0.5 × 6.022×10²³ = 3.011×10²³
Each N2 molecule has 2 atoms:
Atoms = 3.011×10²³ × 2 = 6.022×10²³ atoms
```

**Method = 1** **$6.022\times10^{23}$ atoms = 1**

**Q2(vii). Moles and molecules in 11 g of $CO_2$.** **3 marks**

```
Molar mass of CO2 = 12 + 2(16) = 44 g/mol
Moles = mass / molar mass = 11 / 44 = 0.25 mol           [1½]
Molecules = 0.25 × 6.022×10²³
          = 1.506×10²³ molecules                    [1½]
```

### OR — Balance: $Fe + H_2O$ (steam) → $Fe_3O_4 + H_2$.

```
Balanced: 3Fe(s) + 4H2O(g) → Fe3O4(s) + 4H2(g)
Check → Fe: 3 = 3   O: 4 = 4   H: 8 = 8   ✓
```

**Balancing = 2** **State symbols = 1**

**Q2(viii). Oxidation state of the central atom in $MnO_2$, $HClO_4$, $H_3PO_4$.** **3 marks**

```
(a) MnO2:  Mn + 2(−2) = 0  → Mn = +4               [1]
(b) HClO4: (+1) + Cl + 4(−2) = 0 → Cl = +7        [1]
(c) H3PO4: 3(+1) + P + 4(−2) = 0 → P = +5     [1]
```

### OR — Redox reaction; oxidized/reduced species in $2Mg + O_2 \rightarrow 2MgO$.

A **redox reaction** is one in which **oxidation** (loss of electrons / increase in oxidation state) and **reduction** (gain of electrons / decrease in oxidation state) occur simultaneously. **1**

In $2Mg + O_2 \rightarrow 2MgO$: **Mg** goes 0 → +2 (loses electrons) = **oxidized**. **1**

**Oxygen** goes 0 → −2 (gains electrons) = **reduced**. **1**

**Q2(ix). Corrosion; how painting and oiling protect iron.** **3 marks**

**Corrosion:** the slow, natural eating away (destruction) of a metal by the chemical action of air, moisture and other substances in its surroundings — e.g. the rusting of iron. **1**

*Both methods work on the same principle:* forming a barrier that keeps out oxygen and water.

(a) **Painting:** a layer of paint coats the iron surface and forms a barrier that stops oxygen and moisture reaching the metal, so it cannot oxidise. **1**

(b) **Oiling / greasing:** a film of oil or grease (used on moving/machine parts) covers the surface and keeps out air and water, preventing oxidation. **1**

### OR — Why aluminium resists corrosion (oxide layer).

Aluminium is reactive and reacts quickly with oxygen in air, but this forms a **thin, hard, continuous layer of aluminium oxide ($Al_2O_3$)** on its surface. **1½**

This oxide layer is **impervious and sticks tightly**, sealing the metal underneath so that no further air or moisture can reach it; the metal below is therefore protected from further corrosion. **1½**

**Q2(x). Why a spark is needed to start an exothermic combustion (activation energy).** **3 marks**

Although the combustion of methane is **exothermic overall** (it releases energy), the reactant molecules must first collide and **break their existing bonds** (C–H and O=O) before new bonds can form. **1**

The minimum energy needed to start the reaction is the **activation energy**. At room temperature very few molecules have this much energy, so the reaction does not start on its own. **1**

A spark or flame supplies this initial **activation energy**; once the reaction begins, the heat it releases keeps supplying energy to the surrounding molecules so it continues by itself. **1**

### OR — Define catalyst; industrial example; effect on rate and on ΔH.

**Catalyst:** a substance that increases the rate of a chemical reaction without itself being consumed (it is chemically unchanged at the end). **1**

**Industrial example:** iron (Fe) in the Haber process ($N_2 + 3H_2 \rightarrow 2NH_3$); or vanadium(V) oxide ($V_2O_5$) in the Contact process. **1**

(a) It **increases the rate** by providing an alternative path of **lower activation energy**. (b) It does **not change ΔH** — the overall enthalpy change stays the same. **1**

**Q2(xi). Differentiate physical equilibrium and chemical equilibrium; examples.** **3 marks**

| Physical equilibrium | Chemical equilibrium |
|---|---|
| Balance in a reversible **physical change**; no new substance is formed. | Balance in a reversible **chemical reaction**; new substances are formed. |
| **Example:** water ⇌ water vapour in a closed bottle (evaporation ⇌ condensation). | **Example:** $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (or $2SO_2 + O_2 \rightleftharpoons 2SO_3$). |

**Difference = 2** **1 example each = 1**

### OR — How forward and reverse rates change until equilibrium.

At the **start (moment of mixing)** only reactants are present, so the **forward rate is maximum** and the **reverse rate is zero**. **1**

As reactants are used up the **forward rate falls**; as products build up the **reverse rate rises**. **1**

Eventually the two rates become **equal** — this is equilibrium. After this the rates stay equal and constant; the reaction continues both ways but concentrations no longer change (dynamic equilibrium). **1**

## SECTION C — Long Questions Answer Key (4 × 5 = 20 Marks)

**Q3(i). (a) Electronic configuration of C and S + group/period. (b) Atom with 12p, 12e.** **5 marks**

**(a) Electronic configuration** = the arrangement/distribution of the electrons of an atom in its shells (and sub-shells) around the nucleus. **1**

```
(i)  Carbon  (Z = 6):  2, 4  → Group IV-A (14), Period 2      [1]
(ii) Sulphur (Z = 16): 2, 8, 6 → Group VI-A (16), Period 3    [1]
```

**(b)** An atom with 12 protons and 12 electrons is **magnesium (Mg)**, configuration 2, 8, 2.

- (i) It is a **metal**, because it has only 2 electrons in its outermost shell, which it tends to **lose** (metals are electron donors). **½**
- (ii) It loses 2 electrons to form the ion $\mathbf{Mg^{2+}}$ (charge = **+2**). **½**
- (iii) Its **valency = 2**. **1**

**OR**

**(a) Ionization energy** = the minimum energy required to remove the outermost (most loosely bound) electron from an isolated gaseous atom to form a positive ion. **1**
Sodium's first ionization energy is much **lower** than chlorine's because: Na has only **1 valence electron** and a larger size, and losing it gives the stable neon configuration, so little energy is needed; Cl has a **smaller size, higher nuclear charge (17 protons)** holding its electrons tightly and 7 valence electrons it does not lose easily, so much more energy is needed. **2**

```
(b) Increasing atomic size: Al < Mg < Na                     [1]
Reason: Na, Mg, Al are in the same period (Period 3). Across a
period nuclear charge increases while electrons fill the same
shell, so the nucleus pulls them in more strongly and size
decreases. Na (11 p) largest → Al (13 p) smallest.         [1]
```

**Q3(ii). (a) Empirical/molecular formula; $CH_2O$, mass 180. (b) Molecules in 3.6 g $H_2O$.** **5 marks**

**(a) Empirical formula** = the simplest whole-number ratio of atoms of each element in a compound. **Molecular formula** = the actual number of atoms of each element in one molecule. **1**

```
Empirical formula mass of CH2O = 12 + 2(1) + 16 = 30 amu
n = molecular mass / empirical mass = 180 / 30 = 6            [1]
Molecular formula = (CH2O)6 = C6H12O6  (glucose)         [1]
```

**(b)**

```
Molar mass H2O = 2(1) + 16 = 18 g/mol
Moles = 3.6 / 18 = 0.2 mol                                    [1]
Molecules = 0.2 × 6.022×10²³ = 1.204×10²³ molecules  [1]
```

**OR**

**(a) Mole** = the amount of a substance that contains $6.022 \times 10^{23}$ (Avogadro's number) representative particles; its mass equals the molar mass in grams. **1**

```
(i)  CH4: molar mass = 12 + 4(1) = 16 g/mol
     moles = 8 / 16 = 0.5 mol                                 [1]
(ii) N2: molar mass = 2(14) = 28 g/mol
     mass = 2 × 28 = 56 g                                    [1]
```

**(b)**

```
Balanced: 2Al + Fe2O3 → Al2O3 + 2Fe                       [1]
```

It is a **redox reaction**: Al is oxidized (0 → +3, loses electrons) and Fe is reduced (+3 → 0, gains electrons); oxidation and reduction occur together. **1**

**Q3(iii). (a) Redox in Zn + $2AgNO_3$. (b) O.S. of Mn in $MnO_4^{2-}$, S in $SO_3^{2-}$.** 5 marks

**(a) Oxidation** = loss of electrons (increase in oxidation number). **Reduction** = gain of electrons (decrease in oxidation number). 1

In **$Zn + 2AgNO_3 \rightarrow Zn(NO_3)_2 + 2Ag$**:

- Zn: 0 → +2 (loses 2 electrons) = oxidized → **Zn is the reducing agent**. 1
- $Ag^+$: +1 → 0 (gains 1 electron) = reduced → **$AgNO_3$ ($Ag^+$) is the oxidizing agent**. 1

```
(b)(i) MnO4²⁻: Mn + 4(−2) = −2 → Mn = +6            [1]
(b)(ii) SO3²⁻:  S + 3(−2) = −2 → S = +4             [1]
```

**OR**

**(a) Oxidizing agent** = a substance that gains/accepts electrons and is itself reduced. **Reducing agent** = a substance that loses/donates electrons and is itself oxidized. 1

In **$MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$**:

- Mn: +4 → +2 (gains electrons, reduced) → **$MnO_2$ is the oxidizing agent**. 1
- Cl: −1 → 0 (in $Cl_2$) (loses electrons, oxidized) → **HCl is the reducing agent**. 1

```
(b) Balanced: 2Fe + 3Cl2 → 2FeCl3                          [1]
Iron is oxidized (Fe: 0 → +3).                            [1]
```

**Q3(iv). (a) ΔH for $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$. (b) $N_2 + 3H_2 \rightleftharpoons 2NH_3$.** 5 marks

**(a)** ΔH = (bond energies of bonds broken) − (bond energies of bonds formed).

```
Bonds BROKEN (reactants):
  CH4 → 4 × (C–H) = 4 × 414 = 1656
  2O2 → 2 × (O=O) = 2 × 498 =  996
  Total = 2652 kJ
Bonds FORMED (products):
  CO2 → 2 × (C=O) = 2 × 803 = 1606
  2H2O → 4 × (O–H) = 4 × 463 = 1852
  Total = 3458 kJ
ΔH = 2652 − 3458 = −806 kJ                                [2]
```

Since ΔH is **negative**, the reaction is **exothermic**. 1

**(b)** (i) The symbol ⇌ shows the reaction is **reversible** — it goes both forward ($N_2$+$H_2$→$NH_3$) and reverse ($NH_3$→$N_2$+$H_2$) at the same time. 1

(ii) **No**, the reaction does not stop at equilibrium. The forward and reverse reactions continue at **equal rates**, so the concentrations stay constant while both reactions still occur (dynamic equilibrium). 1

**OR**

**(a)**

```
Bonds BROKEN (reactants):
  H–H = 436 ;  Br–Br = 193  → Total = 629 kJ
Bonds FORMED (products):
  2 × (H–Br) = 2 × 366 = 732 kJ
ΔH = 629 − 732 = −103 kJ                                  [2]
```

ΔH is **negative** → the reaction is **exothermic**. 1

**(b)** (i) **Respiration** = **exothermic**, ΔH is **negative** (energy is released). 1

(ii) **Photosynthesis** = **endothermic**, ΔH is **positive** (energy from sunlight is absorbed). 1

**— End of Answer Key —**